Name_____ Date_____ Period_____

LESSON 6-3 · HOMEWORK PACKET · SHEET 1 OF 2

Writing Repeated Factors as Powers

Maryland Standard 6.EE.A.1 · Learning target – I can write repeated multiplication as a power and name its base and exponent.

REMEMBER

- A **power** shows a repeated factor: $5 \times 5 \times 5 = 5^3$.
- The **base** is the repeated factor. The **exponent** tells how many times it is a factor.
- Read 5^3 as "5 to the third power" or "5 cubed." Read 5^2 as "5 squared."

WORKED EXAMPLE

Write $6 \times 6 \times 6 \times 6 \times 6$ as a power. Name the base and the exponent.

Step 1 The repeated factor is 6. It is the base.

Step 2 Count the factors: there are 5. That is the exponent.

Step 3 Write the base with a raised exponent: 6^5

Answer **6^5; base 6, exponent 5**

PRACTICE

show your work · label units

1. Write each product as a power.

a. $4 \times 4 \times 4 =$ _____

b. $9 \times 9 =$ _____

c. $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 =$ _____

d. $10 \times 10 \times 10 \times 10 =$ _____

e. $1/3 \times 1/3 \times 1/3 =$ _____

f. $0.5 \times 0.5 =$ _____

g. $6 \times 6 \times 6 \times 6 \times 6 =$ _____

h. $2/5 \times 2/5 =$ _____

2. Complete the table.

Power	Base	Exponent	Expanded form
8^3			
	7	4	
			11×11

3. **Phone Tree.** A coach calls 3 players. Each player calls 3 more players, and so on. In round 5, the number of calls is $3 \times 3 \times 3 \times 3 \times 3$. Write this as a power.

Power_____

4. **Find the Error.** Rafael wrote $4 \times 4 \times 4 \times 4 \times 4$ as 5^4. Explain his mistake. Then write the correct power.

_____

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME Explain to someone at home why 2^4 does not mean 2×4.

Name_____ Date_____ Period_____

LESSON 6-3 · HOMEWORK PACKET · SHEET 2 OF 2

Finding the Value of a Power

Maryland Standard 6.EE.A.1 · Learning target – I can find the value of a power by multiplying the base the number of times the exponent shows.

REMEMBER

- Write the **expanded form**, then multiply one factor at a time.
- $4^3 = 4 \times 4 \times 4 = 64$, **not** $4 \times 3 = 12$.
- Fraction and decimal bases work the same way: $(1/2)^2 = 1/2 \times 1/2 = 1/4$.

WORKED EXAMPLE

Find the value of 2^5.

Step 1 Expanded form: $2 \times 2 \times 2 \times 2 \times 2$

Step 2 Multiply: $2 \times 2 = 4$, $4 \times 2 = 8$, $8 \times 2 = 16$, $16 \times 2 = 32$

Answer $\mathbf{2^5 = 32}$

PRACTICE

show your work · label units

1. Find the value of each power.

a. 7^2 = _____ **b.** 3^4 = _____ **c.** 5^3 = _____

d. 10^5 = _____ **e.** 2^6 = _____ **f.** 1^8 = _____

2. Find the value. Write fractions in simplest form.

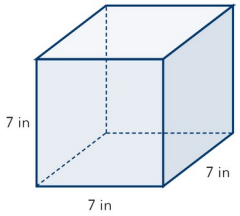
a. $(1/2)^3$ = _____ **b.** 0.3^2 = _____

3. Write each number as a power of the given base.

a. 27 as a power of 3: ____ **b.** 10,000 as a power of 10: ____

4. **Storage Cube.** A storage cube has edges that are 7 inches long. Its volume is 7^3 cubic inches. Find the volume.

Show your work.



Answer_____ in^3

5. **Compare.** Select the **two** powers that have the same value.

☐ **A** 2^6 ☐ **B** 6^2 ☐ **C** 4^3 ☐ **D** 3^4

TEST PRACTICE

choose the best answer

6. What is the value of 4^3?

A 7 **B** 12 **C** 64 **D** 81

7. Which expression has a value of 1,000?

A 10^3 **B** 3^{10} **C** 10×3 **D** 100^3

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask someone at home to guess which is greater, 2^5 or 5^2. Then find both values together.